

RANBIR KUMAR

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PROFESSIONAL SUMMARY

AI-focused software engineer (B.Tech Computer Science, CGPA 8.28) who takes systems from research into production. Shipped an agentic RAG document platform serving real users, a CMS running an industry body's live website behind CI/CD, and a routing app deployed at a national fintech conference across 393 exhibitor stalls; implemented Prototypical Networks from scratch to 93% accuracy on scarce medical labels. Filed patent plus peer-reviewed research in vision perception and explainable AI, with 4 vision pipelines running on NVIDIA Jetson under 40 ms.

EDUCATION

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology, Chennai | CGPA: 8.28/10 | Aug 2023 – May 2027 (expected)

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Explainable AI, DBMS & SQL, Probability & Statistics

EXPERIENCE

Software Testing & Product Development Intern | **IAMAI** (Internet and Mobile Association of India), Delhi | May 2026 – Jul 2026

- Inherited fully manual pre-release validation on two production web platforms and owned automating it: built an **LLM-planned browser agent** (Playwright + Chromium) reading DOM and accessibility snapshots, executing safe UI actions, watching console and network traffic, and emitting machine-readable defect reports. It now generates, executes, and triages regression runs across critical user flows, cutting manual click-through effort an estimated 70% per release, while k6 load, stress, and spike profiles plus Burp Suite scans fixed degradation and vulnerability thresholds before go-live.

Technical Integration Lead, AI Systems & Embedded Products | **IORA**, Noida | May 2025 – Apr 2026

- Hardware, firmware, and application teams were building an nRF9151 LTE-M/NB-IoT safety wearable against divergent assumptions. Owned integration architecture: ran design reviews across PCB, firmware (MQTT over TLS, GNSS, flash-buffered telemetry), and power, converging 3 teams onto one interface contract, and architected the company's agentic AI platform (see Projects), which held availability through LLM vendor outages via failover across 3 providers.

Deep Learning & Computer Vision Intern | **Proeffico Solutions Pvt. Ltd.**, Noida | May 2025 – Oct 2025 (on-site May–Jul; remote Aug–Oct)

- Client sites relied on staff watching live RTSP camera feeds around the clock. Designed, profiled, and deployed 4 production vision pipelines (speed monitoring, pothole, fire, workplace safety) on NVIDIA Jetson Xavier NX, selecting compact detector backbones over heavier variants for a conservatively estimated 60% cut in inference compute, landing sub-40 ms latency (roughly 25 FPS per stream) and replacing continuous manual review with automated real-time alerting.

Rover Lead, Autonomous Systems & Computer Vision | **Technocrats Robotics**, VIT Chennai | Aug 2025 – Aug 2026 (member since Mar 2024)

- Took over a 60-member Rover Department for the 2025-26 season with autonomy split across sub-teams; owned subsystem architecture, technical direction, and the competition build timeline. Delivered the perception stack — CV soil classification, arrow-marker detection, and an end-effector camera pipeline on Raspberry Pi and an i7 mini-PC — carrying the team to All-India Rank 1 in DD Robocon Round 1, 20th worldwide of 100+ teams at IRC 2025, and IRC 2026 Finalist.

PROJECTS

IORA DocQA — Production Agentic RAG Document Intelligence Platform | *FastAPI · Next.js · pgvector · DuckDB · Multi-LLM*

github.com/Ranbirkumar26/iora-docqa

- Teams needed answers out of long PDFs without invented citations, so grounding had to be enforced rather than prompted. Built the ingestion path — parsing, chunking, and embedding generation into a Supabase pgvector store — behind a FastAPI retrieval service with a hard coverage gate that refuses to answer whenever retrieval cannot support the question, trading a slice of answer coverage for effectively zero ungrounded responses in live use.
- Extended it with a DuckDB text-to-SQL path over uploaded CSV/XLSX, persistent per-user memory, and a semantic answer cache resolving an estimated 25% of repeat questions with no LLM call, plus failover across Gemini, Groq, and OpenRouter; six weeks of production hardening behind approval-based signup and admin RBAC kept it serving real users through vendor outages that would otherwise have taken the product down.

IAMAI Publishing Platform — Full-Stack CMS & Content Delivery System | *Next.js · TypeScript · Supabase · Postgres · GitHub Actions*

github.com/Ranbirkumar26/IAMAI_CMS_Public

- An industry body's public website needed a developer for every content change. Designed a normalized Postgres schema spanning eleven content collections with CMS-managed page blocks driving every public page, so non-technical staff now publish events, reports, policy papers, and member listings directly, and routine content updates ship with no code deploy and no engineering involvement at all.
- Implemented Supabase authentication with password reset, user and permission management, and RBAC across CMS routes, then modularized the codebase and shipped it through a GitHub Actions pipeline to Netlify, adding a scheduled smoke-test monitor that verifies the live deployment on a recurring schedule and catches broken releases before anyone checks the site manually.

TerraView — Indoor Venue Navigation & Multi-Floor Routing Engine | *React 19 · Vite 7 · Graph Pathfinding · Netlify* jio-map.netlify.app

github.com/Ranbirkumar26/jio-map

- Global Fintech Fest attendees could not locate exhibitors across a multi-hall, multi-floor venue and signage alone was not solving it. Modelled the whole convention centre as a single routing graph over stall, amenity, gate, and lift-connector nodes so shortest-path search resolves cross-hall and cross-floor journeys, rendering each as guided previous/next segments across 4 floor views at Jio World Centre.
- Layered exhibitor search over 393 mapped stalls and 225 companies, gated by a pre-deploy validation script that blocks release whenever a sold stall lacks company metadata, plus an admin authoring mode for gate and routing edits; ships as an offline-capable static bundle with zero backend and zero hosting cost, embedded directly in native Android and iOS WebViews with no wrapper application required.

RetinaProto — Few-Shot Medical Image Classifier with Grad-CAM Explainability | *PyTorch · Prototypical Networks · ResNet-18 · Grad-CAM*

github.com/Ranbirkumar26/Explainable-Fewshot

- Retinal disease labels are scarce and clinicians will not act on opaque predictions, so the model had to learn from few examples and justify itself. Implemented Prototypical Networks from scratch — episodic N-way K-shot sampling, class-prototype computation, Euclidean-distance inference — over a ResNet-18 encoder trained end to end on hospital-sourced images, reaching 93% accuracy and generalizing to unseen disease classes from a handful of labels.
- Reported seeded episodic evaluation as mean and standard deviation across repeated episodes instead of a single favourable run, and layered Grad-CAM attribution exposing the retinal regions behind each prediction; the resulting work forms the basis of a peer-reviewed journal submission currently under review.

RESEARCH, PATENTS & ACHIEVEMENTS

Patrolling Bot — AI-Powered Autonomous Campus Surveillance Robot | *Patent filed* — replaced fixed-post guard rounds with YOLO detection (missing ID cards, fire/smoke, overcrowding) and action recognition for fight detection, alerting security in real time from Jetson Xavier NX and Raspberry Pi with LiDAR/ultrasonic navigation (Python, OpenCV, PyTorch, ROS) running unattended across campus patrol routes.

Vision-Based Terrain Analysis | *Accepted; publication forthcoming* — fused YOLOv8 detection with MiDaS monocular depth so a rover classifies terrain from one camera instead of added depth sensors, optimized to run on embedded edge hardware.

Achievements: Patent Filed · All-India Rank 1, DD Robocon R1 · IRC Top 20 Worldwide 2025, Finalist 2026 · SIH 2025 Finalist · Insightify 6.0 Finalist

SKILLS

Languages & Software Engineering: Python, TypeScript, JavaScript, Java, SQL; Data Structures & Algorithms, Graph & Shortest-Path Algorithms, Object-Oriented Programming, System Design, REST API Design, Authentication & Authorization (RBAC, RLS), Caching, Performance Optimization

Frontend: React, Next.js (App Router), Vite, Tailwind CSS, shadcn/ui, HTML5, CSS3, Responsive UI, Client State Management

Backend: FastAPI, Flask, Node.js, REST APIs, WebSockets, Server-Sent Events, Microservices, Background Workers (Celery/Redis)

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, ResNet, Prototypical Networks, Few-Shot Learning, YOLOv8, MiDaS, OpenCV, Object Detection, Edge Inference (Jetson Xavier NX, Raspberry Pi), Explainable AI (Grad-CAM, SHAP, LIME)

Generative AI & Agentic Systems: Agentic AI, AI Agents & Autonomous Agents, Multi-Agent Orchestration, LangGraph, Tool / Function Calling, Model Context Protocol (MCP), Agentic RAG Pipeline Design, Grounded Generation, Guardrails & Coverage Gating, Semantic Search, Embeddings, Vector Databases (pgvector, FAISS, Qdrant), LangChain, LlamaIndex, HuggingFace Transformers, Semantic Caching, Text-to-SQL, Prompt Engineering, Local LLM Inference (Ollama)

Databases, Cloud & DevOps: PostgreSQL, Supabase (Auth, RLS, Storage), DuckDB, SQLite, Schema Design, Pandas, NumPy, Docker, Git, GitHub Actions (CI/CD), Linux, Vercel, Netlify, Railway, Render, Playwright, Selenium, pytest, k6